

The Server in the Room: Why Hong Kong Must Regulate AI's Hidden Climate Cost

In 2022 July night in Kowloon, while thousands of air conditioners hum to keep families cool, a few miles away, rows of black server racks are generating enough heat to boil a swimming pool every hour. Outside, Hong Kong was recording its warmest July on record. At least 3 people died from heatstroke in one weekend (Hong Kong Free Press, 2022). The machine built to make us smarter was making the planet hotter. Hong Kong is unknowingly fueling a thermal crisis.

A recent article from Hong Kong Free Press (2026) showed us that global investment in data centres nearly doubled between 2022 and 2024, reaching half a trillion dollars. Locally, the forthcoming Sandy Ridge data center is expected to have a carbon footprint account for 2% of the city's total emissions but the policy governing the emission from data center remains out of date. The Energy Saving Plan 2015–2025 has expired and never addressed data centres at all (Hong Kong Free Press, 2026). At this rate, the ambitious 2050 carbon neutrality target set by Hong Kong government can never be met. Globally, data centre electricity consumption is expected to grow at around 15% per year from 2024 to 2030. This is more than four times faster than the growth of all other sectors combined (International Energy Agency, 2025). The technology widely considered as a solution to climate change has become one of the few sectors where emissions are projected to grow further.

The solution begins with you. Two individual actions create the pressure that moves companies providing AI tools.

First, audit your own AI footprint. Every question you ask AI consumes energy equivalent to leaving a light bulb on for roughly twenty minutes. Tools like Machine Learning CO2 Impact Calculator (<https://mlco2.github.io/impact/>) allow you to estimate your carbon footprint from a stupid question you asked AI online. Share the number to your friends and the community. When individuals make the invisible visible, people will start to understand the environmental impact of AI.

Second, switch to providers who have an acceptable Power Usage Efficiency (PUE) of around < 1.2 and have disclosed their carbon emission. Providers who publish emissions data are more accountable. It creates market pressure that regulation alone cannot generate overnight. Advanced liquid cooling technologies are already present and can reduce energy used for cooling by up to 50%. Also, moving facilities to low-carbon grid regions can halve the combined carbon footprints (Patano et al., 2026). It proved that it is always possible for data centers to improve their environmental friendliness. What is missing is an incentive. In 2023, Singapore ties government subsidies to the Data Center Carbon Footprint Assessment (DC-CFA) program directly to PUE benchmarks, creating a financial incentive. More than 20 companies were “lured” by this incentive and submitted their proposal (Infocomm Media Development Authority, 2023). This proves that market pressure will be very effective in improving a company's policy.

These two actions are the prime examples of how individual behaviour becomes institutional pressure. Individual demand is half of that alignment. The European Union now makes carbon reporting for all data centres above 500 kilowatts across 27 countries mandatory (European Parliament & Council of the European Union, 2023). Hong Kong, however, has no equivalent. Your actions can change this status. In 2022, the warmest July was recorded (Hong Kong Free Press, 2022). Climate change is not a joke and it is already affecting our day to day life. Together we can reduce the carbon footprint and contribute to the global effort to save the environment.

The server in the room cannot speak. You do. Measure your footprint today at mlco2.github.io, understand the PUE and carbon emission of your favourite AI provider, share this essay to your friends and family, and ask one question: what does our AI costs?

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